

Core Java - TOC				
Section	Chapter	Chapter Name	Topic	Training Mode
Java Basics	Chapter-1	Introduction to Java,Data Types and Variables	History & Features of Java: OOP, platform independence, JVM, JDK, JRE Data Types: Primitive and non-primitive types, Variable declaration, Type casting Object-Oriented Principles: Encapsulation, Inheritance, Polymorphism, Abstraction	In-Person Trainer - ILT (ILT - Instructor Led Training) Daily Lab Exercise(DLE)
Object- Oriented Programming (OOP) Concepts	Chapter-2	OOP Basics	Classes and Objects: Class structure, Object creation, Instance variables Constructors: Default constructor, Parameterized constructor, Constructor overloading HandsON : Practice - TekStac Lab	
	Chapter-3	Inheritance,Abstraction & Encapsulation	Inheritance: Single inheritance, Multilevel inheritance, super keyword Polymorphism: Method overloading, Method overriding Abstraction: Abstract classes, Interfaces Encapsulation: Access modifiers (public, private, protected), Encapsulating data HandsON : Practice - TekStac Lab	
	Chapter-4	Keywords	this and super: Referring current object, Accessing parent class methods Static Members: Static variables, Static methods, Static blocks final Keyword: Final variables, Final methods, Final classes	
	Chapter-5	Arrays,Strings,StringBuffer & StringBulider	Arrays: One-dimensional array, Multidimensional array, Array traversal and operations Strings:String class, Immutability, String methods: length, substring, equals, etc. HandsON : Practice - TekStac Lab	In-Person Trainer - ILT DLE & BU Connects
	Chapter-6	Exception Handling	StringBuffer & StringBulider: Mutability, Common methods: append, insert, delete Types of Exceptions: Checked exceptions, Unchecked exceptions try-catch Block: Single try-catch, Multiple catch, finally block throw and throws: Manually throwing exceptions, Propagating exceptions HandsON : Practice - TekStac Lab	
Collections	Chapter-7	Java Collections	Custom Exceptions: Creating custom exceptions Introduction to Collections: Collection hierarchy, Collection interfaces (List, Set, Map) List Interface: ArrayList, LinkedList, Vector, Stack HandsON : Practice - TekStac Lab	In-Person Trainer - ILT DLE & BU Connects
			Set Interface: HashSet, LinkedHashSet, TreeSet Map Interface: HashMap, LinkedHashMap, TreeMap, Hashtable Iterator: Iterator, ListIterator HandsON : Practice - TekStac Lab	
	Chapter-8	Java Thread Handling	Comparable & Comparator: Sorting with Comparable, Sorting with Comparator Introduction to Threads: Thread vs Process, Thread lifecycle Creating Threads: Extending Thread class, Implementing Runnable interface Thread Methods: sleep(), join(), start(), run()	In-Person Trainer - ILT DLE & BU Connects
	Chapter-9	Java File Handling	Character Streams: FileReader, FileWriter, BufferedReader, BufferedWriter HandsON : Practice - TekStac Lab	
JDBC Overview	Chapter-10	JDBC (Java Database Connectivity)	File Handling: File class, File reading and writing (FileInputStream, FileOutputStream) JDBC API Overview: JDBC architecture, Driver Manager, Statement and PreparedStatement CRUD Operations: Insert, Update, Delete, Select Queries Transactions: Commit, Rollback, Savepoints HandsON : Practice - TekStac Lab	In-Person Trainer - ILT DLE & BU Connects
Java 8 Basics	Chapter-11	Java 8 Features	Lambda Expressions: Lambda syntax, Functional interfaces Stream API: Stream operations (filter, map, reduce) Optional Class: Avoiding NullPointerException with Optional	
Advanced Topics	Chapter-12	Generics & Annotations	Generics: Generic classes, methods & wildcards HandsON : Practice - TekStac Lab	In-Person Trainer - ILT DLE & BU Connects
			Annotations: Built-in annotations (@Override, @Deprecated) & Custom annotations	
	Chapter-13	Design Patterns	Design Patterns: Singleton, Factory, Observer, Strategy patterns HandsON : Practice - TekStac Lab	In-Person Trainer - ILT DLE & BU Connects